

Lab 4: Vibe Coding Chat History

Project and lab requirements

Assigned repository: https://github.com/SETAPESU26/08_helicopter

Personal fork: https://github.com/PES1UG24AM369/08_helicopter

The handout requires fixing the starter code, implementing README features one by one, recording a 10-second video before and after changes, pushing the completed code to a personal repository, keeping separate commits per task, and submitting both videos, updated code, and chat history. This is an individual assignment.

Initial setup

1. Forked the assigned repository to the personal GitHub account.
2. Cloned the fork and opened the project in VS Code.

```
git clone https://github.com/PES1UG24AM369/08_helicopter.git
cd 08_helicopter
```

3. Installed dependencies and launched the original game. A 10-second BEFORE video was recorded before codechanges.

```
pip install -r helicopter/requirements.txt
python helicopter/main.py
```

Task 1 — Fix movement and screen boundaries

Prompt used: Inspect the existing helicopter implementation and fix the laggy direction change and screen-boundary bugs. Keep the project structure and controls. Do not implement collision detection, scoring, or shields yet. Explain the files changed.

Implementation summary: Updated the helicopter movement so Up/Down inputs set a predictable vertical velocity, and clamped the helicopter to the top and bottom screen boundaries.

Validation: User confirmed the helicopter stayed within the screen and movement worked.

```
git add helicopter/game/helicopter.py
git commit -m "Fix helicopter movement and boundaries"
```

Task 2 — Collision detection and Game Over

Prompt used: Add collision detection between the helicopter and the top/bottom obstacle walls. Trigger Game Over when either wall is hit, keep the gap safe, and stop game updates after Game Over. Do not add scoring or shield features yet.

Implementation summary: Added a Game Over state and collision checks in GameEngine. The code was adapted to the existing Obstacle API, using `get_top_rect()` and `get_bottom_rect()`. A visible GAME OVER message was added.

Validation: User confirmed collision and Game Over behavior worked correctly.

```
git add helicopter/game/game_engine.py
git commit -m "Add obstacle collision and game over"
```

Task 3 — Distance-based scoring

Prompt used: Add a distance score that increases during gameplay, remains visible after Game Over, and stops increasing when the game ends. Preserve existing movement and collision behavior.

Implementation summary: Initialized self.distance = 0 in __init__() and incremented the value during update() after the Game Over early return. The score was displayed using the renderer.

Validation: User confirmed the score worked perfectly.

```
git add helicopter/game/game_engine.py git
commit -m "Add distance-based scoring"
```

Task 4 — One-use shield

Prompt used: Add a visible, manually activated shield that protects against one obstacle collision. The shield should disappear after absorbing a collision; a subsequent collision should cause Game Over. Preserve the earlier tasks.

Implementation summary: Used Spacebar activation, shield_active and shield_used state, and a visible blue outline. Initial collision handling was refined after testing because the helicopter could remain overlapping the same obstacle after the shield was consumed. The final approach tracks the obstacle absorbed by the shield and ignores that obstacle until it has passed the helicopter.

Validation: User confirmed the shield now worked as intended.

```
git add helicopter/game/game_engine.py git
commit -m "Add one-use shield mechanic"
```

Key debugging notes

- Added import pygame to game_engine.py for pygame drawing and key constants.
- A temporary AttributeError reported that GameEngine had no update method; this was resolved and the game ran again.
- The shield's initial short grace-period approach could allow Game Over while the helicopter was still overlapping an obstacle. Tracking the shielded obstacle proved more reliable.

Testing checklist

- Movement changes direction responsively.
- Helicopter stays within top/bottom screen boundaries.
- Colliding with obstacle walls triggers Game Over; the gap is safe.
- Distance increases during play and stops after Game Over.
- Space activates the shield; one obstacle encounter is absorbed; a later collision triggers Game Over.
- Record and save the AFTER video (approximately 10 seconds).
- Push commits to the personal fork and verify they appear on GitHub.
- Place Before_Video.mp4, After_Video.mp4, updated code, and this PDF in the Lab-4 submission folder.

Git verification and push commands

```
git log --oneline -4
git status git push
origin main
```

Important: This PDF is a structured summary of the work discussed, not a literal transcript of all messages. If the instructor explicitly requires a verbatim ChatGPT export, export the conversation using ChatGPT's available sharing/export controls as well.